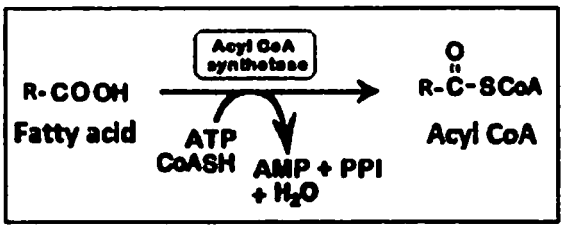
Two processes control the amount of triacylglycerols in depot fat: lipogenesis and lipolysis

II. *Lipogenesis:*

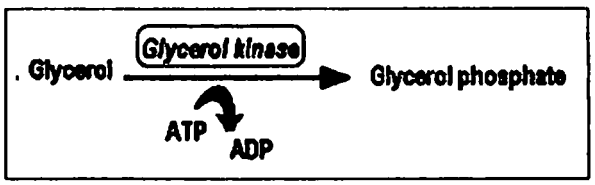
A. **Definition:** Lipogenesis is the synthesis of triacylglycerols from fatty acids (acyl CoA) and glycerol (glycerol-3-phosphate).

1. **Activation of fatty acids into fatty acyl CoA:**



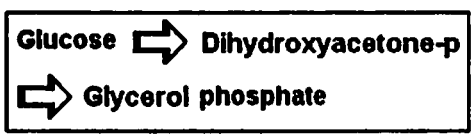
2. **Synthesis of glycerol phosphate :**

a) **In liver, kidney, intestine, and lactating mammary glands:** Glycerol phosphate is formed from glycerol by **glycerokinase** or from glucose through

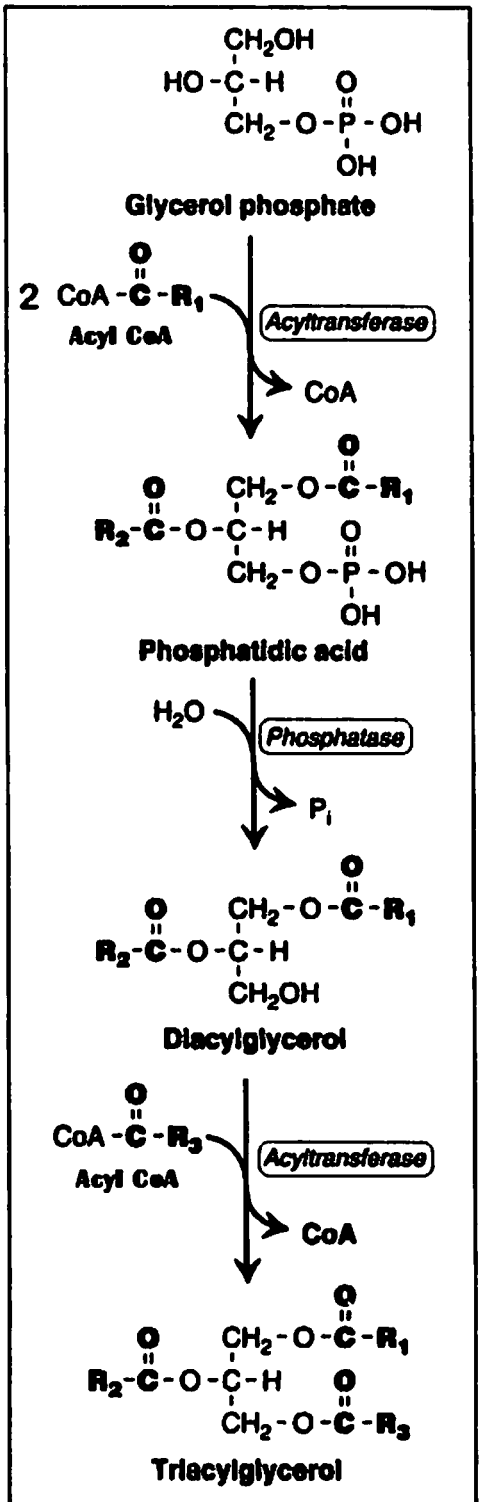


glycolysis.

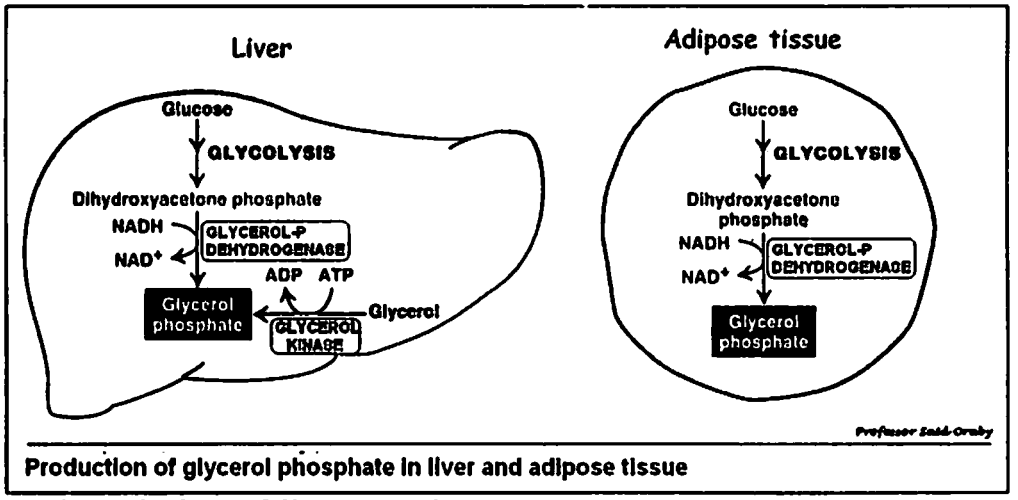
b) **In muscles and adipose tissue, glycerokinase is deficient. In these tissues glycerol phosphate is formed from glucose (through glycolysis) as follows:**



3. **Synthesis of triacylglycerol:** by reaction between acyl CoA and glycerol phosphate as shown in the diagram.



Synthesis of triacylglycerols.
Oraby



Production of glycerol phosphate in liver and adipose tissue

B. Regulation of lipogenesis:

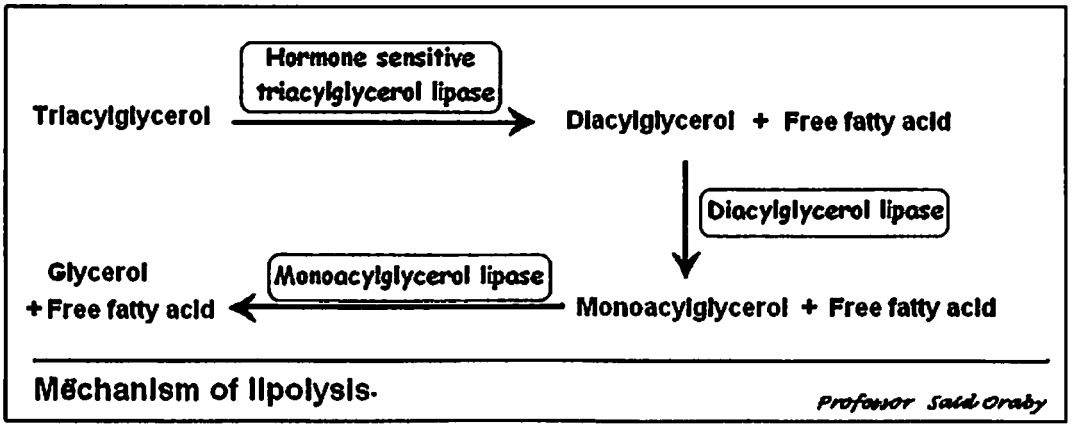
1. **After meal, lipogenesis is stimulated:** Insulin is secreted which stimulates glycolysis. Glycolysis supplies dihydroxyacetone phosphate that is converted into glycerol phosphate in adipose tissue.
2. **During fasting lipogenesis is inhibited,** as anti-insulin hormones e.g. glucagon are secreted. These inhibit lipogenesis and stimulate lipolysis.

III. *LIPOLYSIS :*

A. Definition and location: Lipolysis is the hydrolysis of triacylglycerols into glycerol and fatty acids. Its location is the cytosol of adipose tissue cells.

B. Steps: Lipolysis is carried out by a number of lipase enzymes, which are ***present in adipose tissue.*** These are :

1. Hormone sensitive triacylglycerol lipase.
2. Diacylglycerol lipase.
3. Monoacylglycerol lipase.



C. Fate of products of lipolysis:

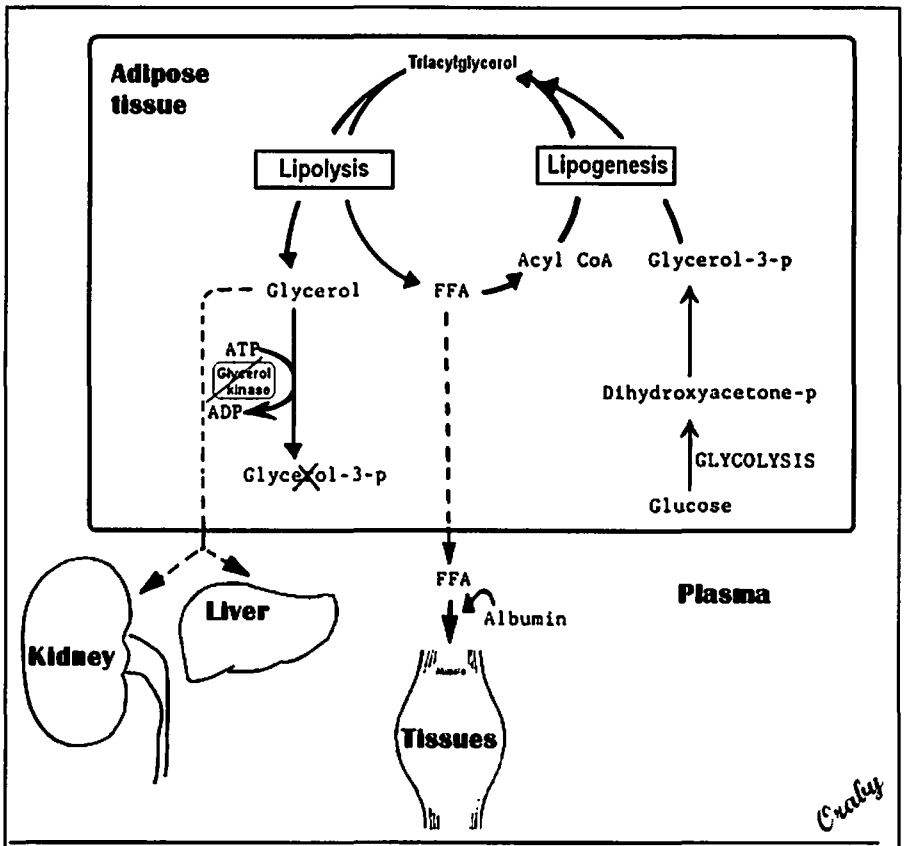
1. **Fate of fatty acids:**
   a) Oxidation by tissues to give energy.
   b) Fatty acids may remain in adipose tissue to be re-esterified into triacylglycerols again.

2. Fate of glycerol:

a) Glycerol may diffuse to blood and then taken up by the liver to give:

1) Glucose by gluconeogenesis.
2) Pyruvate by glycolysis.
3) Triacylglycerols by lipogenesis (outside adipose tissue).

> **Note:** In adipose tissue, glycerokinase enzyme is deficient, which is essential to convert glycerol into glycerol phosphate. Thus glycerol in adipose tissue cannot be used in re-esterification of fatty acids to form triacylglycerols.



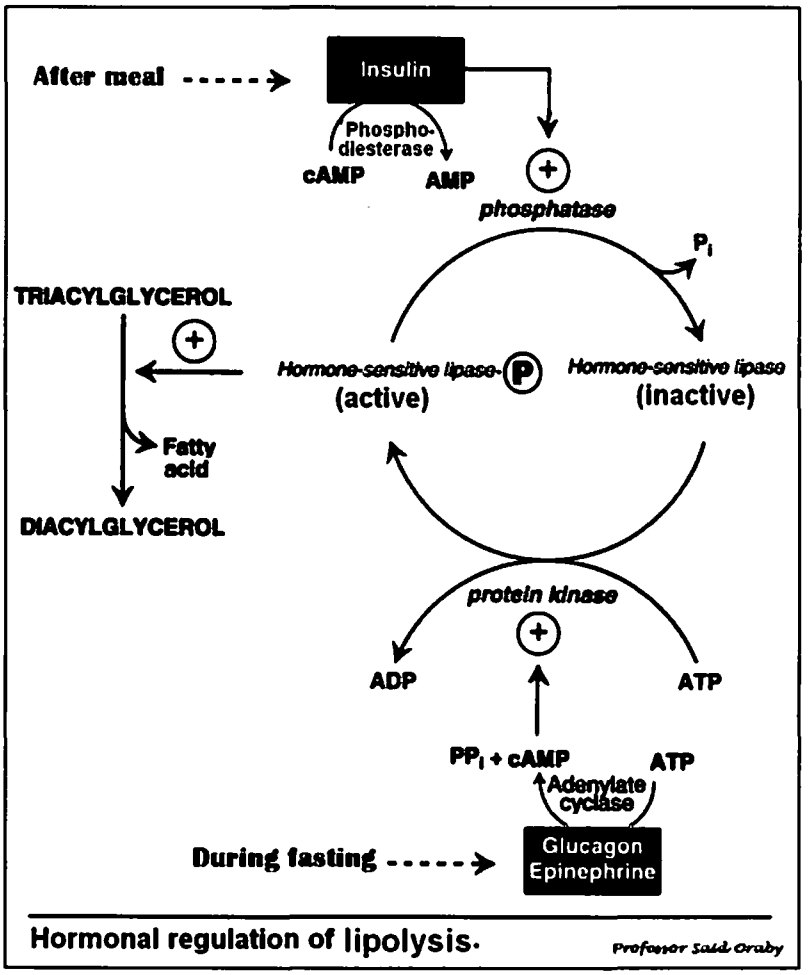
Lipolysis and fate of its products.

D. <u>Regulation of lipolysis</u>: The key enzyme controlling Lipolysis is the hormone sensitive triacylglycerol lipase. It exists in 2 forms: active (phosphorylated) and inactive (dephosphorylated)

1. During fasting:

a) Many hormones (epinephrine, norepinephrine and glucagon) are secreted. They stimulate adenylate cyclase enzyme ➔ formation of cAMP ➔ Activation of protein kinase enzyme ➔ Phosphorylation of hormone sensitive triacylglycerol lipase ➔ Stimulation of lipolysis.

2. **After meal:**
   a) Insulin is secreted ➔ Stimulation of both phosphodiesterase enzyme and lipase phosphatase enzyme ➔ dephosphorylation and inactivation of hormone sensitive triacylglycerol lipase enzyme ➔ Inhibition of lipolysis
   b) Caffeine is a substance present in coffee. It inhibits phosphodiesterase enzyme ➔ Stimulation of lipolysis.



E. **Causes of excessive lipolysis:** In conditions where the need for energy is increased (low insulin and high glucagon):
   1. Starvation.
   2. Diabetes mellitus.
   3. Low carbohydrate diet.
   4. Hypercatabolic states e.g. fevers, hyperthyroidism.